

Answers:

- 1.a) they equal in concentration
b) as you add water, you add equal amounts of OH^- and H_3O^+ , but amount you add is very small compared to incredibly large amount of water added so % of each ion decreases until essentially they are equal and adding water has no more effect since you would be adding equal amounts OH^- and H_3O^+ so you can no longer disrupt the balance
- 2.a) 8.60 b) $6.31 \times 10^{-6} \text{ mol/L}$ c) $6.31 \times 10^{-6}/3 = 2.10 \times 10^{-6} \text{ mol/L}$ d) 5 intervals on log scale 00 000 times more acidic
- 3.a) both determine how dangerous an acid can be or both identify that the solution contains a lot of H^+ ions
b) strong is determined by molecular properties and concentrated is determined by the amount added to water or strong is determined by the extent to which the acid breaks apart into ions and concentrated is determined by total number of acid molecules added to water
c) although the soluble acid dissolves well into water, only a small proportion becomes H^+ ions and although all of the strong acid is capable of becoming H^+ ions, little of it dissolves in water so it only delivers a small proportion of H^+ ions
d) although the weak acid only forms a small proportion of H^+ ions, so much of it was added to water that it could match the pH of the small amount of strong acid which completely becomes ions